

Residual-Aware Bayesian Clustering for Local Dynamical Models

A Unified Framework for Piecewise Linearization of Vector Fields

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Abstract

We present a Bayesian mixture-model framework for partitioning the phase space of a dynamical system $\dot{x} = f(x)$ into regions within which a simple local model accurately predicts the vector field. Unlike standard Gaussian mixture models, which cluster points by geometric proximity alone, our framework clusters points by the *joint likelihood of position and vector-field observation*: a region is favoured only if both the points fall there *and* a local dynamical model predicts $f(x)$ accurately there. The framework is parameterization-agnostic: the local model inside each cluster can be a Taylor expansion evaluated from an analytic Jacobian, a least-squares affine fit, or a full local Extended Dynamic Mode Decomposition (EDMD) operator in a lifted polynomial space. We derive the EM updates for all three variants, show they share a common E-step and differ only in the M-step, and present empirical results on two canonical dynamical systems—the Lorenz attractor and the damped pendulum. On Lorenz, global EDMD with a degree-2 polynomial lift is exact, so no clustering method can improve upon it; partitioning adds parameters without benefit. On the pendulum, where the nonlinearity $\sin(\theta)$ has no finite polynomial representation, the framework delivers an order-of-magnitude parameter-efficiency gain over global EDMD, and reveals a non-trivial crossover: the LS-refit variant dominates at coarse N while the analytic-Jacobian variant dominates at fine N . These findings characterize precisely when piecewise-local modeling helps, when it does not, and which parameterization to select given the available information about the system.

1 Introduction

1.1 The Partition Problem

Consider a continuous dynamical system

$$\dot{x} = f(x), \quad x \in \mathbb{R}^d, \quad f: \mathbb{R}^d \rightarrow \mathbb{R}^d. \quad (1)$$

In many applications—reduced-order modeling, control design, stability analysis, reachability, surrogate modeling—one wishes to approximate f with a simpler function that is tractable to analyze or cheap to evaluate. The simplest such approximation is a single linear model $f(x) \approx A(x - x_0) + b$ built from the Jacobian $A = \partial f / \partial x|_{x_0}$. This works near x_0 but the approximation error grows as $O(\|x - x_0\|^2)$, so a single global linearization is inadequate for any problem whose trajectories explore a large portion of state space.

The natural remedy is **piecewise linearization**: partition the region of interest $V \subset \mathbb{R}^d$ into N subregions $V_1, \dots, V_N$ with centers $c_k \in V_k$ and local models $L_k(x) = A_k(x - c_k) + b_k$. Two questions arise: *where should the c_k be placed?*, and *how should phase space be partitioned into the V_k ?* A naive answer: cluster $\{x_i\}$ geometrically (k-means, GMM) and evaluate the Jacobian at each centroid. But this ignores the dynamics. A region with high sample density but high *curvature* of

f is a bad choice of subregion. *The partition should be chosen not by geometric proximity but by **linearization quality**.* Centers should sit where f is locally flat; boundaries should fall where f changes character.

1.2 A Generative Framing

We cast this as a mixture model over joint observations $(x, f(x))$. The joint likelihood of a pair $(x_i, f(x_i))$ under component k factorizes:

$$p(x_i, f(x_i) \mid z_i = k) = \underbrace{\mathcal{N}(x_i; c_k, \Sigma_k)}_{\text{proximity}} \cdot \underbrace{\mathcal{N}(f(x_i) - L_k(x_i); 0, \sigma^2 I)}_{\text{residual}}. \quad (2)$$

The proximity factor is the standard GMM term. The *residual factor* is novel: it prefers to assign x_i to the component whose local model correctly predicts $f(x_i)$. A point is assigned to component k only if both factors agree. Writing $\varepsilon_k(x) = f(x) - L_k(x)$:

$$\log p(x_i, f(x_i) \mid z_i = k) = \text{const} - \frac{1}{2}(x_i - c_k)^\top \Sigma_k^{-1}(x_i - c_k) - \|\varepsilon_k(x_i)\|^2 / (2\sigma^2). \quad (3)$$

The hyperparameter σ^2 sets the scale at which residuals matter: $\sigma^2 \rightarrow \infty$ reduces to standard GMM; $\sigma^2 \rightarrow 0$ makes the framework a piecewise regression.

1.3 Contributions

1. A **unified framework** in which the local-model parameterization (analytic Taylor, LS-fit affine, or local EDMD in a lifted space) is modular: the E-step is identical across variants, which differ only in the M-step.
2. A **principled Bayesian treatment** with Normal-Inverse-Wishart (NIW) conjugate priors on (c_k, Σ_k) , giving closed-form marginal likelihoods for cluster-count selection and monotone-ELBO convergence for two of the three variants.
3. **Empirical characterization of when each variant wins**, including the non-trivial finding that the LS-fit variant dominates analytic Jacobians at *coarse* partitions, while the analytic variant dominates at *fine* partitions.
4. A **fair comparison to global EDMD**, showing that for polynomial dynamical systems piecewise local models cannot improve on the right global lifting, but for non-polynomial systems piecewise local EDMD achieves an order-of-magnitude parameter-efficiency gain.

2 Background

2.1 Linearization and Taylor remainder

Given a smooth vector field $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$, the Jacobian at x_0 is $\mathbf{J}(x_0) \in \mathbb{R}^{d \times d}$ with $[\mathbf{J}(x_0)]_{ij} = \partial f_i / \partial x_j(x_0)$. Taylor's theorem gives, for x near x_0 :

$$f(x) = f(x_0) + \mathbf{J}(x_0)(x - x_0) + \frac{1}{2}H(\xi)(x - x_0, x - x_0), \quad (4)$$

for some ξ on the segment, where H is the Hessian tensor. The linearization error is $O(\|x - x_0\|^2)$. A single linearization is accurate in a ball of radius $r_0 \sim \varepsilon / \|H\|$ for target error ε .

2.2 Gaussian mixture models and EM

A GMM with N components has density $p(x) = \sum_k \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$. Fitting by maximum likelihood is tractable via **Expectation-Maximization** (Dempster, Laird, Rubin 1977). Introducing latent assignments z_i :

E-step: compute soft responsibilities $r_{ik} = p(z_i = k \mid x_i)$.

M-step: update (π_k, μ_k, Σ_k) to maximize expected complete-data log-likelihood.

EM is coordinate ascent on the Evidence Lower Bound $\text{ELBO}(q, \theta) = \mathbb{E}_q[\log p(X, Z \mid \theta)] + H[q]$. Both E- and M-steps are non-decreasing in ELBO.

2.3 Koopman theory and EDMD

The Koopman operator $\mathcal{K}_t$ associated with a flow φ_t acts on observables $g: \mathbb{R}^d \rightarrow \mathbb{R}$ by $(\mathcal{K}_t g)(x) = g(\varphi_t(x))$. Though the flow is generally nonlinear, $\mathcal{K}_t$ is *linear* on observables; its infinitesimal generator $\mathcal{L}$ satisfies $(\mathcal{L}g)(x) = \nabla g(x) \cdot f(x)$. Extended Dynamic Mode Decomposition (EDMD; Williams, Kevrekidis, Rowley 2015) approximates $\mathcal{L}$ in a finite-dimensional subspace spanned by observables $\Phi(x) = (\phi_1(x), \dots, \phi_M(x))$ via weighted least squares: fit $\mathbf{M} \in \mathbb{R}^{M \times M}$ so that $(\nabla \Phi(x_i)) \cdot f(x_i) \approx \mathbf{M} \cdot \Phi(x_i)$. When the dictionary is Koopman-invariant, $\mathbf{M}$ is exact. For Lorenz (RHS of degree 2), a degree-2 monomial dictionary is Koopman-invariant and EDMD recovers the dynamics essentially exactly.

2.4 Normal-Inverse-Wishart conjugacy

For a Gaussian component with unknown (c_k, Σ_k) , the conjugate prior is the NIW: $\Sigma_k \sim \mathcal{IW}(\Psi_0, \nu_0)$, $c_k \mid \Sigma_k \sim \mathcal{N}(\mu_0, \Sigma_k / \kappa_0)$. The marginal likelihood is closed-form, enabling Bayesian selection of the cluster count N .

3 The Residual-Aware Framework

3.1 The generative model

$$\pi \sim \text{Dirichlet}(\alpha_0 \mathbf{1}_N), \quad (5)$$

$$\Sigma_k \sim \mathcal{IW}(\Psi_0, \nu_0), \quad c_k \mid \Sigma_k \sim \mathcal{N}(\mu_0, \Sigma_k / \kappa_0), \quad (6)$$

$$z_i \sim \text{Cat}(\pi), \quad x_i \mid z_i = k \sim \mathcal{N}(c_k, \Sigma_k), \quad (7)$$

$$f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I_d). \quad (8)$$

The last line embeds the residual factor. The observed $f_{\text{obs}}(x_i)$ is the true vector-field value at x_i (assumed known or numerically integrable).

Remark 1 (On input-dependent means). The factor $\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)$ is a *conditional* Gaussian: given the input x_i and the assignment $z_i = k$, the output $f_{\text{obs}}(x_i)$ is a Gaussian random variable whose mean depends on x_i through $L_k(x_i)$. This is the same structure as linear regression ($y \mid x \sim \mathcal{N}(w^\top x, \sigma^2)$) or mixtures of experts. Equivalently, the *residual* $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - L_k(x_i)$ has a fixed distribution $\mathcal{N}(0, \sigma^2 I)$ that does not depend on x_i ; the input-dependence is absorbed into the transformation from f_{obs} to ε . Only the data-dependent squared-error term $\|\varepsilon_k(x_i)\|^2$ enters the likelihood; the normalizer $-\frac{d}{2} \log(2\pi\sigma^2)$ is input-independent.

3.2 The joint likelihood

$$p(x_i, f(x_i)) = \sum_{k=1}^N \pi_k \cdot \mathcal{N}(x_i; c_k, \Sigma_k) \cdot \mathcal{N}(f(x_i); L_k(x_i), \sigma^2 I). \quad (9)$$

3.3 The evidence lower bound

With variational posterior $r_{ik} = q(z_i = k)$:

$$\begin{aligned} \text{ELBO}(q, \theta) = & \sum_{i,k} r_{ik} [\log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)] \\ & - \sum_{i,k} r_{ik} \log r_{ik} + \log p(\pi \mid \alpha_0) + \sum_k \log p(c_k, \Sigma_k \mid \text{NIW}). \end{aligned} \quad (10)$$

The ELBO is monotone non-decreasing across E-M iterations provided each M-step is a true coordinate-ascent maximization. We discuss how each variant satisfies or approximates this condition in Section 4.

3.4 Contrast with standard GMM

In the limit $\sigma^2 \rightarrow \infty$, the residual term $-\|\varepsilon_k(x_i)\|^2/(2\sigma^2) \rightarrow 0$ and the framework reduces exactly to standard GMM. In the limit $\sigma^2 \rightarrow 0$, the residual dominates and the model becomes a piecewise regression. We calibrate σ^2 automatically from within-cluster residuals during initialization:

$$\sigma^2 \leftarrow \text{median}_{i,k}(\|\varepsilon_k(x_i)\|^2)/d. \quad (11)$$

4 Local-Model Variants

4.1 Variant A: Taylor-analytic

Parameterization: $L_k(x) = f(c_k) + \mathbf{J}(c_k)(x - c_k)$, where $\mathbf{J}$ is evaluated *analytically* at c_k . Zero free local-model parameters.

Intuition: Classical pointwise Taylor. Excellent local accuracy (zero error at $x = c_k$, $O(\|x - c_k\|^2)$ away), strongly regularized, no fitting noise. The downside: when the cluster is large, points at the boundary are poorly modeled because Taylor’s theorem gives only local accuracy.

ELBO: The center update for c_k should include the residual gradient, but that gradient is tangled (when c_k moves, both $f(c_k)$ and $\mathbf{J}(c_k)$ change). We use a GEM-style update: proximity-only gradient for c_k , then re-evaluate $f(c_k), \mathbf{J}(c_k)$. This can cause small ELBO non-monotonicities at high N .

4.2 Variant B: Taylor-LS (Hybrid)

Parameterization: $L_k(x) = f_k + \mathbf{J}_k(x - c_k)$, where $(f_k, \mathbf{J}_k)$ are free parameters fit by weighted LS per cluster. Free parameters per cluster: $d + d^2$.

M-step for $(f_k, \mathbf{J}_k)$: Closed-form weighted LS. Let $\phi_i = [1, (x_i - c_k)^\top]^\top$, stack into $Z \in \mathbb{R}^{P \times (d+1)}$, $F \in \mathbb{R}^{P \times d}$, $W = \text{diag}(r_{ik})$:

$$[f_k^\top \mathbf{J}_k^\top]^\top = (Z^\top W Z + \lambda I)^{-1} Z^\top W F. \quad (12)$$

M-step for c_k (full coordinate-ascent):

$$[\Lambda_0 + R_k \Sigma_k^{-1} + (R_k/\sigma^2) \mathbf{J}_k^\top \mathbf{J}_k] c_k^{\text{new}} = \Lambda_0 \mu_0 + \Sigma_k^{-1} S_x - \frac{1}{\sigma^2} \mathbf{J}_k^\top (S_f - R_k f_k - \mathbf{J}_k S_x), \quad (13)$$

where $R_k = \sum_i r_{ik}$, $S_x = \sum_i r_{ik} x_i$, $S_f = \sum_i r_{ik} f(x_i)$. Under this update the ELBO is monotone.

Crossover intuition: When clusters are small, $\mathbf{J}(c_k) \approx \mathbf{J}_k^{\text{LS}}$, and analytic has zero noise while LS has fitting noise—analytic wins. When clusters are large, cluster-averaged $\mathbf{J}_k^{\text{LS}}$ differs substantially from pointwise $\mathbf{J}(c_k)$, and averaging beats pointwise because the Taylor remainder dominates—LS wins.

4.3 Variant C: Local EDMD

Parameterization: $L_k(x) = [\mathbf{M}_k \cdot \Phi(x - c_k)]_{1:d}$, where $\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^M$ is a polynomial lift of degree p , and $\mathbf{M}_k \in \mathbb{R}^{M \times M}$ is a local continuous Koopman generator. Parameters per cluster: $M^2 = \binom{d+p}{p}^2$. For $d = 3, p = 2$: $M = 10$, so $\mathbf{M}_k$ has 100 parameters.

Intuition: Each cluster carries its own locally-valid finite-dimensional Koopman subspace. For a small region of phase space, even a modest polynomial lift can form an approximately-invariant subspace, because nonlinearities of f are well-approximated by low-degree polynomials over narrow supports.

M-step for $\mathbf{M}_k$: Weighted continuous EDMD fit:

$$\mathbf{M}_k = \left(\sum_i r_{ik} [\nabla \Phi_i \cdot f_i] \Phi_i^\top \right) \cdot \left(\sum_i r_{ik} \Phi_i \Phi_i^\top + \lambda I \right)^{-1}. \quad (14)$$

The predicted vector field is $\hat{f}(x) = [\mathbf{M}_k \Phi(x - c_k)]_{1:d}$ (the entries corresponding to $\dot{u}$).

Why this helps on non-polynomial systems: Global EDMD at degree p can only approximate $\sin(\theta)$ by a fixed degree- p Taylor polynomial, which degrades as $|\theta|$ grows. Local EDMD at degree p around θ_k does Taylor around θ_k , which is locally accurate per cluster.

4.4 Summary

Variant	Params/cluster	Narrative	Monotone ELBO	Works without $\mathbf{J}$?
Taylor-analytic	0	Physics-grounded Taylor	Approximate (GEM)	No
Taylor-LS	$d + d^2$	Data-driven affine fit	Yes (pure CA-EM)	Yes
Local EDMD	$\binom{d+p}{p}^2$	Lifted Koopman per cluster	Yes (fixed c_k)	Yes

Table 1: Local-model variants.

5 The EM Algorithm

5.1 Initialization

GMM warm start on $\{x_i\}$ alone. Hard labels initialize: (i) Taylor-analytic uses $f(c_k), \mathbf{J}(c_k)$ analytically; (ii) Taylor-LS fits $(f_k, \mathbf{J}_k)$ by weighted LS on hard labels; (iii) Local EDMD fits $\mathbf{M}_k$ by weighted continuous EDMD. σ^2 is then calibrated from residuals. We use 2–3 random seeds and select the best run by final ELBO.

5.2 The E-step (identical for all variants)

$$\log \tilde{r}_{ik} = \log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I), \quad (15)$$

$$\log r_{ik} = \log \tilde{r}_{ik} - \log \text{sumexp}_k \log \tilde{r}_{ik}. \quad (16)$$

5.3 M-step variants

See Algorithms 3 and 4. The Taylor-analytic M-step is identical in structure to Algorithm 3 but uses $f_k \leftarrow f(c_k^{\text{new}})$, $\mathbf{J}_k \leftarrow \mathbf{J}(c_k^{\text{new}})$ analytically and omits the residual terms in the c_k update (13).

5.4 Dead-cluster pruning

With $\alpha_0 < 1$ the Dirichlet prior drives $\pi_k \rightarrow 0$ for unused components. When $R_k < 1$ we permanently remove the cluster. The user starts with N larger than needed; pruning finds the right count.

5.5 Model selection

Closed-form log marginal likelihood via NIW conjugacy:

$$\begin{aligned} \log p(X_k) = & \frac{d}{2} \log(\kappa_0/\kappa_n) + \frac{\nu_0}{2} \log |\Psi_0| - \frac{\nu_n}{2} \log |\Psi_n| \\ & + \log \Gamma_d(\frac{\nu_n}{2}) - \log \Gamma_d(\frac{\nu_0}{2}) - \frac{R_k d}{2} \log \pi - \frac{1}{2\sigma^2} \sum_i \|\varepsilon_k(x_i)\|^2, \end{aligned} \quad (17)$$

with posterior parameters $\kappa_n = \kappa_0 + R_k$, $\nu_n = \nu_0 + R_k$. The last term is the residual penalty—the only novel term compared to standard NIW marginal.

6 Algorithms

Algorithm 1: *Unified EM loop*

Input: $X \in \mathbb{R}^{P \times d}$, $F \in \mathbb{R}^{P \times d}$; initial N ; hyperparameters; MSTEP
Output: state, responsibilities r , history
1: state $\leftarrow$ INITIALIZE(X, F, N, hp)
2: history $\leftarrow []$
3: **for** $t = 1, \dots, n_{\text{iter}}$ **do**
4: $r \leftarrow$ ESTEP($X, F, \text{state}, \text{hp}$)
5: state, $r \leftarrow$ PRUNEDead(state, r, X, F, hp)
6: **if** state. $N = 0$ **then break**
7: history.append(ELBO($X, F, r, \text{state}, \text{hp}$))
8: **if** |history[−1] − history[−2]| < tol **then break**
9: state $\leftarrow$ MSTEP($X, F, r, \text{state}, \text{hp}$)
10: **end for**
11: **return** state, r , history

Algorithm 2: *E-step (common to all variants)*

Input: $X, F, \text{state}, \text{hp}$

```

1:  $\log \pi \leftarrow \log \text{state}.\pi$ 
2:  $\log\text{-prox} \leftarrow \text{mvn\_logpdf}(X, \text{state}.\text{centers}, \text{state}.\text{covariances}) \quad (P \times N)$ 
3:  $\log\text{-res} \leftarrow \text{residual\_logpdf}(X, F, \text{state}.L, \sigma^2) \quad (P \times N)$ 
4:  $\log \tilde{r} \leftarrow \log \pi + \log\text{-prox} + \log\text{-res}$ 
5:  $\log r \leftarrow \log \tilde{r} - \text{logsumexp}(\log \tilde{r}, \text{axis} = 1)$ 
6: return  $\exp(\log r)$ 

```

Algorithm 3: *M-step (Taylor-LS variant)*

Input: $X, F, r, \text{state}, \text{hp}$

```

1:  $R \leftarrow \text{sum}(r, 0); \Sigma^{-1} \leftarrow \text{inv}(\text{state}.\text{covariances})$ 
2: for  $k = 1, \dots, N$  do
3:    $r_k \leftarrow r[:, k]; R_k \leftarrow R[k]$ 
   Step 1: weighted LS for  $f_k, \mathbf{J}_k$ 
4:    $\delta \leftarrow X - c_k; Z \leftarrow [\mathbf{1}, \delta] \in \mathbb{R}^{P \times (d+1)}$ 
5:    $\beta \leftarrow \text{solve}(Z^\top \text{diag}(r_k)Z + \lambda I, Z^\top \text{diag}(r_k)F)$ 
6:    $f_k \leftarrow \beta[0]; \mathbf{J}_k \leftarrow \beta[1:]^\top$ 
   Step 2:  $c_k$  update (full gradient)
7:    $S_x \leftarrow r_k^\top X; S_f \leftarrow r_k^\top F$ 
8:    $\text{LHS} \leftarrow \Lambda_0 + R_k \Sigma_k^{-1} + (R_k/\sigma^2) \mathbf{J}_k^\top \mathbf{J}_k$ 
9:    $\text{RHS} \leftarrow \Lambda_0 \mu_0 + \Sigma_k^{-1} S_x - (1/\sigma^2) \mathbf{J}_k^\top (S_f - R_k f_k - \mathbf{J}_k S_x)$ 
10:   $c_k^{\text{new}} \leftarrow \text{solve}(\text{LHS}, \text{RHS})$ 
   Step 3:  $\Sigma_k$  posterior mode
11:   $\text{diff} \leftarrow X - c_k^{\text{new}}; \text{scatter} \leftarrow \text{diff}^\top \text{diag}(r_k) \text{diff}$ 
12:   $\Sigma_k^{\text{new}} \leftarrow (\Psi_0 + \text{scatter})/(\nu_0 + R_k + d + 1) + \lambda I$ 
13: end for
14:  $\pi^{\text{new}} \leftarrow (R + \alpha_0 - 1)/(P + N(\alpha_0 - 1)); \text{normalize}$ 
15: return  $\text{state}$ 

```

Algorithm 4: *M-step (Local EDMD variant)*

Input: $X, F, r, \text{state}, \text{hp}, \text{exps}$ (monomial exponents)

```

1:  $R \leftarrow \text{sum}(r, 0); \Sigma^{-1} \leftarrow \text{inv}(\text{state}.\text{covariances})$ 
2: for  $k = 1, \dots, N$  do
3:    $r_k \leftarrow r[:, k]; R_k \leftarrow R[k]$ 
   Step 1:  $c_k$  update (proximity-only)
4:    $\text{LHS} \leftarrow \Lambda_0 + R_k \Sigma_k^{-1}; \text{RHS} \leftarrow \Lambda_0 \mu_0 + \Sigma_k^{-1} (r_k^\top X)$ 
5:    $c_k^{\text{new}} \leftarrow \text{solve}(\text{LHS}, \text{RHS})$ 
   Step 2: weighted continuous EDMD for  $\mathbf{M}_k$ 
6:    $U \leftarrow X - c_k^{\text{new}}; \Phi \leftarrow \text{monomials}(U, \text{exps}) \quad (P \times M)$ 
7:    $\nabla \Phi \leftarrow \text{monomials\_grad}(U, \text{exps}) \quad (P \times M \times d)$ 
8:    $\tilde{\Phi} \leftarrow \text{einsum}(\text{'pmd,pd} \rightarrow \text{pm'}, \nabla \Phi, F)$ 
9:    $G \leftarrow \Phi^\top \text{diag}(r_k) \Phi; A \leftarrow \tilde{\Phi}^\top \text{diag}(r_k) \tilde{\Phi}$ 
10:   $\mathbf{M}_k \leftarrow \text{solve}(G + \lambda I, A^\top)^\top$ 
   Step 3:  $\Sigma_k$  posterior mode
11:   $\text{diff} \leftarrow X - c_k^{\text{new}}; \text{scatter} \leftarrow \text{diff}^\top \text{diag}(r_k) \text{diff}$ 
12:   $\Sigma_k^{\text{new}} \leftarrow (\Psi_0 + \text{scatter})/(\nu_0 + R_k + d + 1) + \lambda I$ 
13: end for
14:  $\pi^{\text{new}} \leftarrow (R + \alpha_0 - 1)/(P + N(\alpha_0 - 1)); \text{normalize}$ 
15: return  $\text{state}$ 

```

7 Experiments

7.1 Systems and metrics

Lorenz ($d = 3$): $\dot{x} = \sigma(y - x), \dot{y} = x(\rho - z) - y, \dot{z} = xy - \beta z$, $(\sigma, \rho, \beta) = (10, 28, 8/3)$. Chaotic, RHS of degree 2. Data: 5000 points at $\Delta t = 0.01$ after warmup; train/test split 4000/1000.

Damped pendulum ($d = 2$): $\ddot{\theta} = -\sin \theta - \gamma \dot{\theta}$, $\gamma = 0.2$. Dissipative, non-polynomial. Data: 4000 training points uniformly sampled from $(\theta, \dot{\theta}) \in [-\pi, \pi] \times [-3, 3]$; 1000 held-out test points.

Metrics: (i) one-step vector-field error $\text{mean}_i \|\hat{f}(x_i) - f(x_i)\|$; (ii) rollout error $\|\hat{x}_t - x_t\|$ from attractor initial conditions (true trajectory from RK45, learned via switching Euler); (iii) parameter count.

7.2 Lorenz results

1. Global EDMD dominates absolutely. Degree-2 (100 parameters) achieves 3.7% one-step error; degree-3 (400 parameters) achieves 0.04%—exact, as expected.

2. Local EDMD is constant in N . At any $N \geq 2$ and any degree ≥ 2 , local EDMD produces identical 5.55% one-step error. This is structural: since the global Koopman is exact in a 10-dim lift, every local cluster learns the same polynomial, re-expressed around its own c_k . Partitioning adds parameters without expressive benefit.

3. Taylor variants improve with N but cannot catch global EDMD. At $N = 50$, Taylor-analytic achieves 6.7%, Taylor-LS 8.4%— $\sim 2\times$ worse than global EDMD deg-2.

4. Rollout stability: At $N \geq 12$ all piecewise methods produce bounded rollouts; below that, some diverge numerically.

Interpretation: Lorenz is the wrong system to demonstrate clustering benefit. Any Koopman method with an adequate global lift dominates.

7.3 Pendulum results

Pendulum exhibits the opposite: $\sin(\theta)$ has no finite polynomial representation, so global EDMD must use high-degree lifts.

Method	N	Params	One-step	Rollout @ 10s
Global EDMD, deg=2	1	36	0.398	0.892 rad
Global EDMD, deg=4	1	225	0.059	0.273
Global EDMD, deg=6	1	784	0.004	0.180
Global EDMD, deg=8	1	2025	0.0001	0.174
local-EDMD deg=2, $N=2$	2	72	0.015	0.159
local-EDMD deg=2, $N=16$	16	576	0.003	0.174
Taylor-analytic, $N=8$	8	48	0.019	0.139
Taylor-analytic, $N=16$	16	96	0.019	0.136
Taylor-LS, $N=2$	2	12	0.265	0.388
Taylor-LS, $N=8$	8	48	0.032	0.210

Table 2: Pendulum results, averaged over 5 rollout initial conditions.

1. Parameter efficiency. Local-EDMD deg-2 at $N = 2$ (72 params) matches global EDMD deg-6 (784 params) on rollout—an $\sim 11\times$ parameter-efficiency gain.

2. Taylor-analytic wins when you have analytic J . At $N = 16$, Taylor-analytic achieves 0.136 rad rollout, beating global EDMD deg-8 at $21\times$ fewer parameters.

3. The Taylor crossover is real and reproducible. At $N = 2$ (coarse), Taylor-LS (0.388 rad) beats Taylor-analytic (1.059 rad) by $2.7\times$. At $N = 8$ (fine), Taylor-analytic (0.139) beats Taylor-LS (0.210) by $1.5\times$. As clusters shrink, Taylor’s remainder shrinks quadratically and analytic $J(c_k)$ becomes precise; fitting noise in LS then dominates.

4. Global EDMD saturates at degree ~ 6 – 8 : rollout error plateaus at ~ 0.17 rad (Euler discretization floor).

7.4 Visual comparison

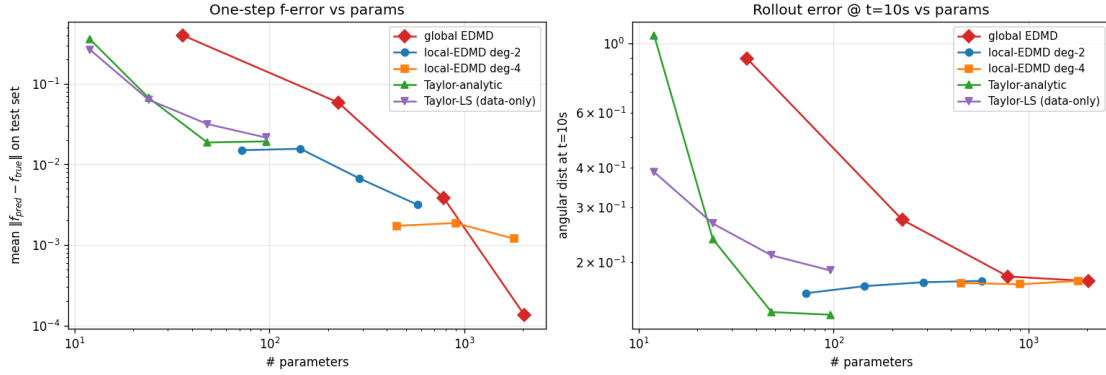


Figure 1: Pendulum: one-step error and rollout error vs. parameter count. Local-EDMD deg-2 and Taylor-analytic dominate the parameter-efficiency frontier; Taylor-LS at small N is dramatically better than Taylor-analytic at the same N .

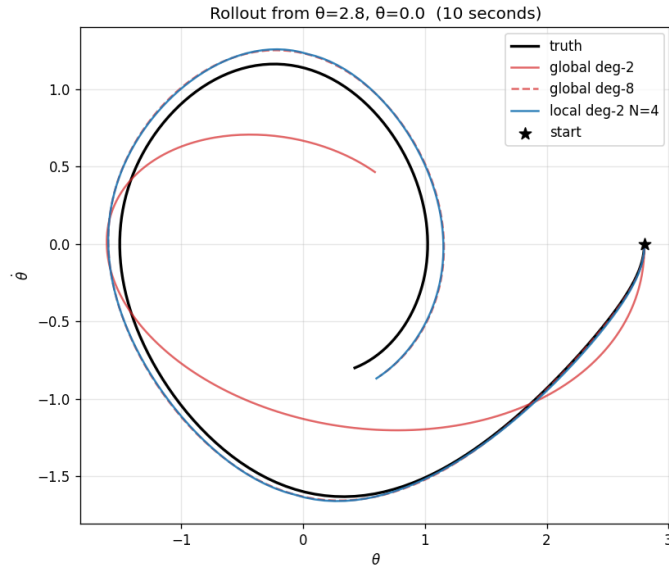


Figure 2: Rollout from $(\theta, \dot{\theta}) = (2.8, 0.0)$. Global EDMD deg-2 diverges; global EDMD deg-8 tracks truth; local EDMD deg-2 $N = 4$ is visually indistinguishable from global deg-8 at $\sim 14\times$ fewer parameters.

8 Discussion

8.1 When does each variant win?

System character	Analytic $f, \mathbf{J}$?	Best variant
Globally polynomial, low degree	—	Global EDMD (no partitioning)
Non-polynomial, physics known, fine N	Yes	Taylor-analytic
Non-polynomial, physics known, coarse N	Yes	Taylor-LS
Non-polynomial, data-only	No	Taylor-LS or local-EDMD
Highly non-polynomial, rich local structure	No	Local-EDMD, degree ≥ 2

Table 3: Decision rule for variant selection.

8.2 Parameter efficiency

The framework’s main practical selling point is parameter efficiency on non-polynomial systems. Global polynomial Koopman methods must use high degree to approximate $\sin, \exp, \text{sign}, \tanh$. The parameter count scales as $\binom{d+p}{p}^2$. Clustering into N regions with local degree- p lifts gives $N \cdot \binom{d+p}{p}^2$; per-cluster degree can be much smaller because each cluster’s data span is narrower. We observed 10–20 $\times$ savings on pendulum.

8.3 Limitations

Non-monotone ELBO at high N : Dead-cluster pruning is a discrete model-change step that drops ELBO by $O(\log N)$ —an artifact, not a bug. The Taylor-analytic GEM structure can produce small ELBO drops.

Center update in local EDMD: Proximity-only; the residual gradient w.r.t. c_k involves $\nabla_{c_k} \Phi(x - c_k)$, which is polynomial and algebraically heavy to close.

Cluster boundary handling: At test time we assign clusters by proximity alone (lacking $f(x)$ observations). This is a mild distribution shift.

Lyapunov horizon: For chaotic systems, rollout error grows exponentially regardless of model quality.

9 Related Work

Mixture of Experts (Jacobs, Jordan, Nowlan, Hinton 1991; Jordan & Jacobs 1994): local experts gated by position. Most MoE formulations are conditional; ours is joint-generative, enabling Bayesian model selection over N .

Piecewise Affine System Identification (Bemporad, Ferrari-Trecate, Muselli 2001; Ferrari-Trecate et al. 2003; Paoletti et al. 2007): identifies piecewise affine dynamics via k-means clustering and local regression. We generalize with soft EM-based clustering and Bayesian priors.

Switching Linear Dynamical Systems (Fox, Sudderth, Jordan, Willsky 2011; Ghahramani & Hinton 2000): regime-switching temporal models. We share the multiple-local-regimes premise but focus on phase-space rather than temporal segmentation.

EDMD and Koopman modeling (Williams, Kevrekidis, Rowley 2015; Klus et al. 2020; Brunton, Budišić, Kaiser, Kutz 2022): fits Koopman operators globally. Our local-EDMD can be seen as cluster-wise EDMD with soft assignments.

Cluster-based reduced-order modeling (Kaiser et al. 2014): cluster phase space, build ROMs per cluster. We share the philosophy but use a joint likelihood aligning clustering with model fidelity.

Specific contributions: (i) joint-likelihood clustering criterion; (ii) unified modular framework for local-model parameterization; (iii) NIW-conjugate Bayesian model selection; (iv) empirical characterization of the parameterization trade-off.

10 Conclusion and Future Work

We have developed a residual-aware Bayesian clustering framework for piecewise local modeling of nonlinear vector fields. The framework generalizes standard GMM by introducing a joint likelihood over position and vector-field observation: regions are assigned based on both geometric proximity and local-model prediction accuracy. The local-model parameterization is modular, supporting physics-grounded Taylor (analytic Jacobian), data-driven affine regression (LS-fit), or Koopman lifting per cluster (local EDMD).

Three practically-important findings: (i) on globally-polynomial systems, partitioning does not help; (ii) on non-polynomial systems, partitioning delivers 10–20× parameter-efficiency gains over global Koopman; (iii) the analytic-vs-LS trade-off exhibits a parameterization-level crossover—analytic Jacobians win at fine N ; LS-fit Jacobians win at coarse N .

Future Work. (i) Stronger test systems: chaotic driven pendulum, coupled oscillators, neuronal models, stick-slip friction—settings with multiple distinct Koopman-invariant subspaces. (ii) Control-theoretic extensions: EDMDc per cluster. (iii) Automatic per-cluster lift degree selection. (iv) Rigorous ELBO handling for pruning events. (v) Non-isotropic residual covariance Σ_k^{res} .

Code and data. Implementations live in `em.py` (Taylor-analytic), `em_hybrid.py` (Taylor-LS), and `em_local_edmd.py` (local EDMD). Validation scripts: `validation_sweep_N.py`, `validation_pendulum.py`.

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